ActInf Textbook Group ~ Cohort 5 ~ Session 2 (Chapter 1, Part 1) ~ 9/26/2023

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all right hello everyone it is cohort five meeting two and we're in our first discussion of chapter one so to begin with what is anyone's just First Take or on chapter one or where does anybody want to begin and then we'll look at the specific questions but is there any just first overviews or just pieces that people want to share about chapter one or just this first discussion you can just go for it if you want to ask me so my biggest um question is around surprise and what the I guess the parameters and boundaries of that is well sorry I think uh a part of my long time persisting uh the literature is like the use of some terms like these and these might be because I'm also a non-native speaker but like the term of like a normative principle like still not sure what it means or like you know later on they say like epistemic this and stuff and I'm just like very confused by those uh terms uh so maybe clarifying yeah what it means to be a normative principle will be a very useful thing for me okay let's just see if it's been asked overall because as Ollie can probably attest to let's just like at least think of oh does anyone want to add a thought on this like just on uses of terms overall or we can look more specifically at what is meant by normativity like in chapter one foreign was also having some um uncertainty about the use of the term I had pre-released the I had an understanding that normative uh the term described something that uh the way that some something should be as opposed to being descript a descriptive theory for example which is which is simply stating what is observed but I uh in the way that it's introduced here normative is used more I think uh placing an emphasis on the ability to to look at a quantity and measure some behavior in terms of that quantity and um uh emphasizing that's better Behavior or behavior that's more in alignment with with the theory is suggesting will will optimize that quantity uh that's my understanding so far but I'd like to be more clear thank you Ali go for it yeah so normative um I mean the use of the the term normative uh here in active inference literature is closely related to the meaning of this term in uh say philosophical literature and stuff uh so uh basically in uh philosophy or more specifically in ethics when we talk about the normative Behavior or um I don't know a normative action uh it's uh it's interpreted as something ethical uh or moral and the reason we ascribe this kind of meaning to normal to this term normative is because we have a standard of action or behavior against which we can compare our own action and then judge by way of comparing it whether it is ethical or not so by the same token we can talk about normativity here in this context of active inference when we talk about a standard for optimization of our action and perception and namely the free energy against which uh the uh the error or the deviation of the action and perception from that standard can be evaluated and then corrected or accounted for so that's why uh I mean researchers in active inference area use the term normal normative or normativity to refer to this kind of compare comparison between the comparison between the action and a kind of a standard for optimization of that action I would kind of question whether that would be tied to ethics though because Norms don't necessarily align with ethical behaviors and of course ethical is subjective but you know um Norms have been you know for example been leaning more toward coercive and exploitive behaviors which in my in my world is not ethical but uh you know say I don't let I'd like to see the you know those two tied apart especially as far as how people act because ethics are you know are not are not Universal um and coerced coercive behaviors uh tend to you know get more get more benefits in U.S society but maybe not other places yeah I guess it's not like so used to me the connection with that it's you know it comes to these ideas from like uh searching first principle Voice or understands uh self-organization of main components so my feeling is that you know there's nothing about our subjective ethical more or more things here so some so yeah I mean the fact that we all have this uh confusion really reveals that maybe normally shouldn't be the work that this thing is here because it's just like well we have so many meanings in so many contexts so I don't know it is very cultural but perhaps I mean if we look at it as just a standard for optimization within a certain a certain definition of a system then you know if we don't get too caught up on our whatever our conceived Notions or subjective Notions of normative as in normal and social norms but just look at it as a optimal Behavior within the parameters of the system hmm yeah so um if I may um clear that up a little bit well uh I mean what I meant it was not that uh uh the the word normativity is used in exactly uh the same way as it's used in I don't know philosophical literature or ethics uh it's just um I mean a related term that's used in its technical sense in active inference literature so I agree that we can maybe uh it's too naive to take that same meaning from philosophical literature and use it directly in this context uh to refer to something I don't know uh as a social Norm or a standard or whatever but uh again in this technical context um Norm here just means a standard against which the optimization of the system happens right so uh that's what I'll wear there is to it so uh it's not something uh it's not something that can be applied to derive any kind of Ethics or morality or whatever uh it's just a way of describing a system uh against a specific standard for optimization namely uh the free energy so that's why this term is used here and also if we want to uh use the first principle way of describing things we need to have these kinds of uh precise uh precise definitions for those standards for optimization uh because uh it's a way of rigorously building up uh our descriptive model so that's necessary to have a very strict and rigorous definition for any standard if we want to build up models rigorously enough to be useful for modeling any complex behavior of um cognitive agents so basically if I understand from this discussion can I you know when in part 1.4.1 says active inference is a normative framework to characterize base optimal behavior and cognition in living organisms does that mean that what it's saying is that active inference as a theoretical framework just sets the objective function of what systems must do which is to minimize the original free energy so normative just refers to the fact that active influence establishes the rules by which systems have to play kind of thing is is that a good interpretation of what normative means yeah more or less that's what all there is to it yeah and also there's a more explicit definition of this term normativity in I guess notes one from chapter one and Page uh let me see uh so yeah here the term normative means there is some evaluative standard against which behavior can be scored active inference is normative in the sense that perception and action are scored by free energy a quantity we will unpack throughout this in the next chapters these are all really great points one analogy that I find very useful or a set like more machine learning and more classically statistical is by now the variational free energy playing the role of the L2 regression fitting Norm in linear modeling so linear modeling doesn't just mean straight lines so check out the earlier friston fmri work on SPM which extends generalized linear modeling so fitting the L2 Norm doesn't mean that the model is adequate it doesn't mean it's going to work if something changes about the underlying data process and so on like it's none of no modeling process can say anything about how well the outcome works but this loss function the L2 norm and linear regression or like the loss function in uh training a neural network these are analogous to the variational free energy such that however you set up a generative model it's going to have this like base optimal like it it's it's going to look like it's on its path of least action given how it's set up with this loss function so I agree with all the absolutely it's very it's a more of like a methodological or a computational normativity in terms of being variational free energy being more like a loss function okay that's one that's one angle on the normativity but I think this is also one discussion that we returned to many times which is just how does something that can describe anything like we describe things to the ex to the extent of their thingness for some statistical pattern or for all these other ways of modeling which is kind of what this um approach ends up looking like in generating these cognitive models like how how can it be measuring everything what is what is something that measures everything say what is the uh meter stick say so understanding like measurement and then all this season so so that's that's an interesting um um concept so it's it's not pointing toward um the outcome as much as it is the value the optimized value would that be in other words you know or is the model you know I understand it's ideology free um or you know I'm assuming I'm I'm inferring that um but you know you the the perception is uh aiming toward a goal so the goal being a value versus an outcome there's a lot there but it it there's just and of course these are just like very various different people's perspective on it but in one view it has equal ideological status to a measurements tool as a measurement tool the kind of narrow sense which it actually looks like when we get through chapter six the recipe for making the active inference generative model which was like alluded to earlier like yeah that's the difference between having a generated model once the generative model is specified slash to the extent it's specified it is going to do the most statistically likely thing but that can include a noisy distribution for perception action so something being uncertain doesn't mean that it's not the likeliest path that is done in the model with like a parameter that describes how noisy something should be that's like variance estimation Bayesian hyper parameters so just because it seems it it shouldn't feel constraining in terms of um developing like a phenomena that seeks information or something like this normativity isn't a like far-reaching claim and then there then there's all of these other kind of aspects that come into playing the actual modeling mostly from the system of Interest just like people use linear modeling across fields and then there's statistics into all of the development that happens in statistics just that that's one kind of deflationary view that is more like the statistical parametric mapping earlier for instant work um and also one other point uh perhaps worth mentioning here is that uh fpp and active inference uh is employed to describe any uh complex system or any self-organizing system um that looks as if it takes uh the action through which um the free energy or that Norm I was talking about is minimized or optimized uh regardless of whether the agent of Interest has any conscious awareness of its intentions or not so it can be applied both to conscious cognitive agents as well as to uh I don't know anything that's that only has uh very simple basal cognition and uh very simple sense of agency and goal-directed behavior so that makes fep an active inference broad enough to Encompass all of these systems um under a similar or even an identical mathematical technology yeah thank you Ali just like the linear regression can be used across different skills the system all the work is in the modeling or so it's this is like the textbook is um oriented towards people on the first half who want to learn about the topic also this is the first textbook so it's kind of like we bring a lot of other things that have happened since and people can continue to bring things in because it's just one view but the first five chapters are more on learning active inferences then the second half even though we don't have all the kind of pieces in place but making the generative model the second half a lot of the very philosophical very general questions it becomes easier to partition the questions into like what's a question more about an implementation in a given system of Interest and then what is the question about the more General um usage of the term but to kind of return all the way to the initial point which is certainly felt by all like the active inference ontology which is to say like the language and the terms like is some terms that are used uh commonly but used in seemingly a different sense here's one way that through the active inference ontology we like at least start to make sense of this and and you're always welcome to use the at sign to like connect it to an ontology term I don't know if every single word but it benefits to see it that way um but like it helps emphasize that it isn't an infinite slash open-ended set of terms that are being used um it's from a few subfields where terms are being used exactly and specifically and technically mostly from Bayesian statistics there's also some terms from on the inbound and on the sense making side from like signal processing ecological psychology as well as on the outgoing more action and control theory so there's those kinds of terms that describe like broadly cybernetic agents are being used essentially exactly and then for many of the terms these are just proposed just which may or may not be used exactly By Any Given author but a lot of the terms have like a narrower technical sense and then a more General usage like attention or anticipation or normativity it's like it like can be used in seemingly different or um a narrower broader way there's a lot to say in in how we all I guess learn and uh use the terms correctly another piece to add just on the ontology is um the terms are not just mostly drawn from these computational areas but the the I I always look at that anyone else want to add something oh just on that following that path uh I'd like to clarify a few things so when we're talking about surprise we're using that term essentially in in the terms of like entropy in information Theory more or less like the Shannon entropy correct and slash if so when we're talking about because minimizing surprise turns out to be challenging for technical reasons you know we use variational free energy as a proxy um so I just like a little Clarity on those technical reasons that minimizing surprise is so difficult is that because it's just too broad of a series of possibilities for a Bayesian calculation um and then so what's the and then what's the mechanism of using free energy as the principle if it's because we're just we're doing a Divergence that's my understanding but clarification would be nice yeah Ali go for it uh yeah we'll come to these questions um uh in chapter two I guess uh and particularly in uh on page 19 and uh 20. uh so uh yeah the idea here is uh although uh the value that needs to be optimized exactly is that the free energy but in practice uh optimizing or minimizing this value precisely or uh in its exact uh Bayesian inference way uh it would prove intractable so the way to somehow solve this solve this issue of intractability is to provide a kind of lower bound that needs to be optimized called variational free energy so uh except for very simple scenarios in every real-time uh where a real-time a real real world situations and there is this um lower bound that needs to be optimized and not exactly the free energy or the surprise although the theory of active inference uh evolves around this idea of optimizing um just a specific normal value namely free energy but uh in practice uh that's the variation of free energy that needs to be um optimize as the lower bound of surprise so yeah a surprise or surprisal is used and it's a statistical sense in active inference literature and it doesn't exactly map onto the psychological sense of the term uh and um again it's a technical word and it doesn't necessarily refer to it doesn't require any cognitive or conscious agent for this kind of informational or statistical surprisal to be defined it's just uh yeah something that can be defined purely in its mathematical terms using the parameters of the system of um yeah inks right so we come from Linguistics so I plan to learn all this math very soon but my question is this um the low road and the high road are they like the same oh you guys can see because I've blurred are they like the same Journey from two points of view like a tube which is a microscope if you look on one side and a telescope if you look on the other side or are they completely traversing different domains the way like syntax and semantics they do meet and they kind of like interlock but they are they they walk the chess board with a different Stitch step Manuel thanks um just before uh I guess addressing that I wanted to say how I understand the limitations so I took to Jesse some I don't know if I'm able am I able to share my screen so I have a presentation I've been planning to give this to my to my research group so I have sort of like visuals to to explain at least the way I understand why minimizing free energy it's it's a it's a good alternative so if is it okay if I would sure how long is the presentation no it's like three slides just to share the concept of why integrating like why Computing all right surprisingly go for it very very fast yeah go for it so basically can you see my slides yep Yeah so basically you know we're in the in the task of computing based ethereum so Computing some latent variable data given in our observation sex and this always involves an integral of this form so the denominator of of Base theorem uh can be computed basically integrating the numerator through all of the possible and unobserved uh uh or latent variables and the way you will do this computationally if you're if most of these integrals don't have a closed form solution so you cannot solve them analytically so what you do numerically is you would imagine you have like a grid and you evaluate so you don't know the height of this function and what the way you're going to integrate is by dividing the grid into different points and then at each point evaluating the height and the integral is going to be basically sort of like based and hot times height so you're going to add all of these small contributions but if you look at this example that I drew here most of the evaluations don't contribute at all to the to the integral because they're all at zero but you don't know that to begin with so evaluating this numerically will become incredibly challenging and furthermore the more the dimensionality of your problem uh increases like the more Dimensions you have the more variables you're trying to infer the more and more and more grid points you will have to evaluate in a so it grows exponentially so it just becomes in in high dimensional spaces evaluating these would become just basically impossible numerically so what we propose is to say okay what I really want is my posterior distribution latent variables given my observations and I'm gonna propose a good enough approximation so this is my true belief that I want to get and this is my proposed close enough so what we'd say is like okay I want my close enough approximation to be as close as possible as a true model and that is done by minimizing these scale but you know that minimizing that distance or pseudometric already involves the distribution that I don't want to work with the one that is really hard to compute but then you do some math that you'll see in the book I guess and you arrive at this that the free energy is uh a lower bound on what we want to compute so basically I don't know how to compute the right hand side of this equation but I know that if I make the left hand side of mass equation as small as possible I would have also made the other one as small as possible given my approximation so so that's how I understand it and there's different yeah I guess that's that's good enough for what I want to say just to shed some light on the technicalities of actually implementing these things computationally uh it just you need a proxy because there's too many dimensions to handle awesome cool yeah very um relevant it's discussed a lot in live stream 52 series with Lance DeCosta some of the relationships with like sampling based methods variational methods all these things um and yeah um Susan and then Rob Moore so yeah that was very helpful um it kind of speaks to the requisite variety aspects I guess um and and that kind of connects to in terms of belief updating and learning um yeah I know that there's variation that is more in the moment more emergent but the and that there's a time lapse in the belief updating but would um would that mean that the goal of learning and updating is widening those boundaries foreign widening um yeah the the requisite choices yeah there's a lot to say there it is related to the cybernetics the principle of requisite diversity and good regulator theorem and those are discussed a lot in the chapter as it's kind of like um like app and then uh often these these three scales of inference um for for so as Alia emphasized there's like continuity with the simple or simplest systems like things that don't have these like higher cognitive functionalities at all um things that do only trivial belief updating like even the case of a ball on the ground you could think of it as doing inference and doing nothing and being at equilibrium so it's a measurement framework that on one hand you can have it totally non-cognitive or basically cognitive like ball rolling down a hill truly as doing some path of least action that's exactly from mechanical physics and the Bayesian mechanics extends that ability to model like mechanical systems and then introduces the faster level of inference and then attention which is precision modulation and then learning which is the slower parameter updating that's within one time within one scale of of uh time and then nested models could Implement arbitrary other learning schemes like here's just another a switch like higher variable that just switches between these knees so that's not learning in the sense if those parameters might be fixed so it's like a narrower statistical learning definition versus like broader human learning so again that's another term like learning and inference and updating or they're not being applied to the human psychological usage it like learning is used in statistics to describe parameter updated why I'm interrupting so but this model does um allow for human learning and I guess would human learning be or you know if it doesn't say so now but uh um and are there model models around policy in terms of you know if the if the goal is human learning I didn't want to speak to that all right go for it so um I don't have an answer to your question Susan so I'm going to uh ask another question and maybe if someone has a comment uh for you they can think about it in the meantime uh my question is related to something that's called Kristen touches on a bit in the preface and then it's it's discussed um another summaries of the free energy principle that you could find for example Wikipedia and it's the uh falsifiability of the free energy principle and a distinction that you might make between a framework or in a principle versus a a theory that's making testable or uh measurable predictions uh that would allow you to confirm or reject a provide evidence for or reject a theory and I would call Kristen and and Thomas Parr balutu in the book is that the free energy principle is is just that it's a principle akin to the theory of evolution and and somehow I haven't been able to make uh sense of what what it is about the free energy principle or or the theory of evolution as a more a common uh Theory or framework that people are more familiar with what is it about those the principles or Frameworks that makes them hard to falsify or not classifiable and then they go on to specify that when these Frameworks do become testable is is when you introduce the generative model it's an active inference the core of the of the Theory comes to to producing a generative model producing some description of what um causes uh in in the worlds generates uh could could be generating the observation is that you're you're seeing and and then that's something that you can go and measure maybe in a human brain or or look for observations to to test against but uh so the essence of my question is what uh distinguishes the free energy Principle as a framework that isn't falsifiable and how does a generative model converted the free energy principle or active inferences to a theory that can be tested great question Ali go for it uh so the reason free energy principle is described as um as an unpalsifiable principle is because it's basically a variational variational principle uh which we use to we use as a mathematical Tool uh to describe the kind of theory [Music] accounts for the behavior of uh particular kinds of self-organizing systems or agents but if we look at this issue in a much broader perspective uh the same problem of falsifiability applies equally to predictive coding Frameworks as well so the question around is predictive coding theories are for false the Bible or not has been going around for several decades now but a very Illuminating uh paper published just a couple of days ago namely is predictive coding falsifiable by Brown by Bowman at all in which they argue that although it looks as if these kinds of theories as predictive coding or active inference and so on looks as if they are falsifiable but the fall survival sorry but the the unfalcifiability of these theories is nothing inherent to these theories at all uh it's just the problem of um Precision that needs to be accounted for if we want to look at the problem of falsifiability for these theories so their conclusion is that these this kind of theories is predictive coding or predictive processing and active inference can indeed be viewed as falsifiable Theory at least in theory so it's nothing not there isn't any inherent barrier for describing these theories as unfalcifiable there isn't any fundamental on falsifiability for these theories but then again when we uh if we come back to active inference again there are two parts for um for the theory of active inference the underlying variational principle namely pre-energy principle and then the empirical Theory cell namely the active inference so even if the underlying principle or fep is unfalsifiable it doesn't necessarily apply to active inference as well so uh active inference can be viewed as a perfectly empirical testable and falsifiable Theory but there are some arguments that the underlying principle uh being as variation for a principle is in essence on falsifiable so yeah thank you ollie there's a lot there um you can always follow where I am in Dakota by clicking on the icon to like improve the notes which would be really helpful or add more information but in this interview that was where friston provided this response that's one answer to this kind of question another is like again to connect it to the linear regression like the statements about variational free energy bounding surprise minimization and then surprise minimization being equivalent to model evidence maximization they are two sides literally of the exact same thing if you have the optimal model you have the lowest bounded surprise lowest surprise that's the exact Bayesian results it's like you're you're doing the best that you possibly could have on the inbound sense making side and then on the outbound control site so again a deflationary statistical description which is accurate of variational free energy is it being a unified tractable bound to surprise minimization whereas there isn't attractable analogous bound directly to model evidence maximization which is why there's the whole question about hill climbing algorithms and and model maximization but this understanding between minimizing something is like equivalent to maximizing something is commonly used like minimizing the training loss on a neural network or minimizing the L2 Norm like the sum of squares on a linear regression it as was mentioned by someone earlier like that doesn't mean that you made the generative model adequate to the real world in any way it's just a statement that the statistical model will have a certain um ability just to process the data routinely like a t-test does that mean that that was an ideology for a t-test or something that that has always been the question that is a separate question than the relationship between variational free energy and surprise which is just a technical finding that it can't be falsified it's just a construction made but then once something is proposed like as a structural hypothesis like in the SPM days they were testing does uh presenting a then B influence this response pattern and then testing structural hypotheses in the brain active inference kind of elaborates on this idea of creating structural hypotheses that themselves being part specific or particular and specified or just evidently not it'd be analogous to asking about like an unspecified linear regression whether you'd expect this or that result let's look at a few other of the questions can I quickly make a comment about uh Russ a question I recently read this paper the math is not the territory navigating the energy principle but I um mayor Andrews and you know my takeaway from that paper which I highly recommend it's that the way to think about the 300 principle is more like uh statistic yeah like a statistical uh framework like you don't falsify calculus calculus is uh you know mathematical framework but if you apply Newton's laws and let's say you have an alternative hypothesis to Newton's clause and they don't agree with the data then those laws are wrong but calculus is still right like calculus is still the mathematical framework to analyze you know differential equations so in that sense uh the 300 principle I think sort of place an analogous role where now the falsifiable thing will cause us then I was saying like you build your generative model maybe you think about Neuroscience you think about how the brain connects or maybe you think about I don't know biophysics how genes talk to each other so in that that is what is going to be falsifiable on the on the free energy principle it's just a way to look at that genitive uh in a in a way to make sense of it like uh what you're proposing kind of thing so so that's my understanding might take away message from from this paper uh yeah yeah great can you please uh link the paper in the yeah live stream 14 also it was discussed with melon with others but yeah that's kind of like a key reference point to help clarify some of this like map territory and the map territory fallacy and the map territory fallacy fallacy we discussed this actually earlier in the cohort for discussion if you wanted to look but let's look at our last like little bit to some of these written questions while the fpp is applicable across many time scales how applicable is active inference as a process theory instantiation of the fep very much plays into our discussion do organisms need a nervous system for it to uh make sense to describe them as doing active inference no you don't it can include inert systems passive systems this isn't the psychological account but it can be applied to mind as a system of Interest but the bazel Machinery that is described in this textbook um like what chapter 4 describes generative models of are not psychological accounts of humans their statistical examples so now there's no need for any kind of specific material or to have to to be material at all it could say oh it's a matrix that just flips between two states it could be as simple as that or you could be applying it just like another analytical or measurement or modeling tool to a system of Interest so and how how applicable is active inference that's the empirical question I'm hoping to get intuitive understanding of how free energy represents a measure of the discrepancy between the organism's internal model of the world and its sensory inputs what is the underlying relationship between information and energy what is the relationship between this free energy and active inference and the thermodynamic concept of free energy that's also present in any organism environment anyone have a thought on this what's the relationship between variational free energy or expected free energy the thermodynamic concept of free energy and I say share my thoughts on that yeah go for it yeah so my understanding is the following so you know in the industrial revolution people are trying to improve uh steam engines and they counted the realization that doesn't matter you know how good your design is how well you know that as perfect as you as your aging because you put some energy but you don't get the the same amount of work out of it there's always some loss there's always something that you cannot really utilize so that leads to Klaus used to make the statement of like oh let's just come up with a concept that we're going to call entropy which is always increasing and it's kind of like we this part of of the of the energy that I am putting to my system that I wasn't able to to make use of like it's just dissipating you can think of friction on other things that just make the the the the all of the energy that you put into the system not you not been utilized later on boltzmann comes in and then he says look entropy is not mysterious entropy is just a measure of how many ways you can arrange a system how many different you know if I have the gas molecules in in this room uh nothing in the loss of physics uh prohibited them to certainly like you know have some random trajectories and collecting that corner and then I'll be suffocating because I'm in a vacuum kind of nothing in the laws of physics for bits that but it's just incredibly unlikely it's just there's so many many many more ways to have them uniformly spread throughout the room that you'll have to wait longer than the age of the universe to see that random event of having them in that corner so now when you uh Shannon comes in and he um formalizes the idea of of uh information as a mathematical quantity that you can put a number on turns out that the metric that he uh derives is exactly of the same functional form as Boltzmann's entropy so you know the two equations look exactly the same and The Story Goes that I think Von Neumann told Shannon oh just use the word entropy that is the same equation and nobody understands the other one either so an advantage you'll win so then later on it I think is EP Jane So a very famous and important physicist in all of these uh statistical mechanics and its uh relationship with based in statistics that I think finally makes a connection between thermodynamic and and informational entropies in that case where uh from my readings and interpretations of what Jane said is that thermodynamic entropy is something I can measure I measure the amount of heat dissipated by my system at a constant temperature and I put a number that's my entropy the information entropy is not something I measure like that it's it's in units of bits but it has to do with in a system if I push a piston how much information I or how many degrees of freedom I control of that system well I can push all of the atoms in the gas and compress them but I don't really have control over their particular trajectories so the work that I get out of that effort has to do with how many uh degrees of freedom I actually control if I was able to go into the atomic level and point all of the gas molecules into the same direction my piston will be much stronger and like push and give me more work but I don't have access to those degrees of freedom so the information I have access to uh is limited therefore if I were to reproduce that piston pushing and uh one or over and over again it will give me the same result given my limited information so I don't know if that's helpful you know it's uh it's uh it's a long uh explanation but uh that's my current understanding of how these two things are connected thanks yeah Kristoff go for it I just want to say something present this from a slightly different angle this is something that um I I took away from my undergrad um I particularly studied uh uh two kinds of semantics of programming languages denotational semantics and operational semantics which is you know you take a program fragment and you're trying to assign a meaning to it and the operational semantics is some kind of sequence of reductions using big or small steps that tells you like you know how the program evolves over time and what the sort of State transitions of the of the virtual machine kind of are and there's a different way of assigning semantic to to a program is to say well there is a mathematical object um that this program represents a good example that would be for instance something like like an infinite list it's not actually realizable in in a finite computer and yet we have programs that act as if they were in infinite lists so we're really hit me is specifically in this in this case was that um you know if something has the common presentation of the of the issue of entropy in uh information Theory versus thermodynamics is that people often mentions like oh these two things have the same mathematical form but there are two different things and I would say that because these things have the same mathematical form there must be the same thing they must be the same thing um it's just the the the depth and the nature of the relationship may not be precisely understood by us just yet but if the mathematics suggests that this it's precisely exactly the same phenomenon because it has the same mathematical form this is kind of a more philosophical uh approach thank you Kristoff another angle so that's a philosophical kind of uh angle uh one is to say A variation for energy is an information theoretic free energy that has that resonance and form or the analogy and form and then when we apply this to self-organizing systems then they if they if we look at things that are existing already they are representing a small um like they're unlikely in one way but then they're likely in their own way so perhaps one way to see it is it's about finding what makes it have a thingness which is what Ali alludes to in the chat with uh fpp and a theory of everything again thing here is used in Iraq that are specific ontological sense definitely we can return to this like well and it's not many times with like this notion of particular States and a particular thing more to say but um

Susan um yeah that was that was a great explanation manual and what piques my interest and I just want to put a pin in uh what you talked about was um the um the correlation between that and the constructal law and yeah I think that that's you know as far as how I kind of tease that apart is is is um the confusion over who designs and I think you know that uh that has a bearing in you know in our updating belief updating so that'd be interesting to pursue I don't know that it's covered in this course but it would be an interesting exercise outside of it cool yeah cool guys quickly Point people if um there's a great great paper um what is information by let me find in my uh David do you remember the author it's just that's I think uh if I if I find it I'll paste it on the on the chat but that just explains for these ideas of entropy you know entropies in the eyes of the beholder like the entropy of a coin is not one bit if I only fire Focus only on the faces yeah it's one bit I just can't have either heads or tails just one yes no question but what if not I not only care about the face of the coin but the angle when I toss it like this is like heading this way or this way and I can discretize it into four corners or eight corners or as many as I want so the question of what is the entropy of the system it's really a matter of who's asking the question and what they want to answer in a sense uh so I thought that is a great point and the relationality is a theme that comes up again and again like just the fundamental octave role of the interactance like the passive Observer and the kind of receptive filter in only like all these ways that the active inference helps reformulate that relationality like even in the colleges that you're mentioning and then the strongest statement I believe um to I mean to christoph's point uh of going further like like the Overton window on this side I think is Alex Kiefer in some of these discussions where he's like it is for like it is further that it is informational but but the the um the actuality of biological systems makes it so that this theoretical construct has like a stronger than um instrumental relationship with thermodynamic free energy like if you had the best foraging algorithm your colony is going to exist in the future the best foraging algorithm doesn't mean the most it just you know means given the niche so that's a normativity that has like a survival normativity to things that survive so that's another piece of how it relates to today Migos yeah it sounds like there might be some unifying concept here that both in the rmodynamic concept there's an energy gradient that has some potential utility for some agent with some goal and in the information there's also from some specific perspective the potential for harnessing that there's if that makes sense it's like if you take a specific vantage point of an agent within that system you can harness information or energy gradients to some end result yeah that's very resonant with Chris fields and Mike Levin's work on like poly Computing which is that the computational attributes are themselves relational so like if something is doing a totally encrypted computation it can't be distinguished from noise but that's like kind of like a um but then encryption is something that is procedural whereas so that has a different set of formalisms than like a statistical distribution the the those are all really good points and ones that that that connect very provocatively but not you know directly necessarily to some of these Felts ways of thinking about that which um yeah interesting times and discussions does anyone want to like add any other comments in the last minutes well hope that was a fun first chapter discussion we'll have the earlier time next week people can add more questions it add and and update the the answers that they're finding relevant um often we have like yeah more people's General Reflections in the first of the discussions and then try to address any and all that because that makes us think about so many things after these discussions that people can just add them here and uh yeah any other comments or thoughts or we'll end it I just want to say there's some very enlightening discussion towards the end and uh thank you all for the contributions yeah thanks a lot guys thank you everyone thank you bye thank you Daniel